

Yash Patel

patel.yashm@northeastern.edu | +1(437) 933-8287 | Toronto, ON | [LinkedIn](#) | [GitHub](#)

EDUCATION

Northeastern University

Master of Science in Bioinformatics

Toronto, ON

January 2024 - December 2025

L.M. College of pharmacy

Bachelor Of Pharmacy

Ahmedabad, India

September 2019- May 2023

SKILLS

Programming & Scripting: Python (scikit-learn, pandas, NumPy, Bioconductor), R (DESeq2, glmnet, ggplot2), SQL, Bash/Shell scripting, Linux/Unix, HTML

Bioinformatics & NGS Tools: FastQC, MultiQC, Trimmomatic, CutAdapt, Seurat, Scanpy, STAR, HISAT2, Salmon, kallisto, GATK, bcftools, BLAST, BEDTools, MAFfilter, FRESCO, HyPhy, clusterProfiler, samtools

Data Analysis & Machine Learning: Differential expression analysis, variant calling (SNP/indel detection), PCA, UMAP, hierarchical clustering, elastic net regression (glmnet), feature selection, dimensionality reduction, FDR correction, correlation analysis, multi-omics integration, biomarker discovery

Genomic Data Formats: FASTA, FASTQ, BAM, VCF, BED, GTF/GFF, MAF

Computing & Workflow Management: High-Performance Computing (HPC), SLURM job scheduling and optimization, Singularity, Docker, Nextflow, GitHub, Generative AI tools (ChatGPT, Claude, GitHub Copilot, Gemini)

Visualization & Reporting: R (ggplot2), Python (matplotlib), data visualization, genomic coverage maps, scientific writing, R shiny

Databases: NCBI, Ensembl, Consensus Coding Sequence Database (CCDS), Uniprot, Genbank

PROJECTS

Bioinformatics Portfolio Dashboard (<https://yashpatel-bioinformatics.streamlit.app/>)

May 2026 – June 2026

- Developed and deployed a live multipage Python and Streamlit portfolio application on Streamlit Community Cloud.
- Integrated Google Gemini 2.5 Flash API to power an AI chat interface for project queries.
- Maintained full version control via GitHub and configured UptimeRobot monitoring to ensure continuous application uptime.

Comparative Genomics: Synonymous Accelerated Elements (120 Species)

September 2025 – December 2025

- Built and automated an end-to-end comparative genomics pipeline using Nextflow on HPC/SLURM, processing 120-species alignments across ~1,980 gene-level datasets to investigate synonymous acceleration with deeper phylogenetic sampling than prior studies.
- Converted genome-wide multiple alignments to gene-level FASTA datasets, and following experimental design executed FRESCO sliding-window analysis, and scanned synonymous substitution rates across 300 genes with optimized SLURM resource allocation, batch scheduling, and result validation.

Bulk RNA-Sequencing: Multiple Myeloma–Adipocyte Crosstalk & Biomarker Discovery

January 2025 – April 2025

- Developed a RNA-sequencing pipeline using Singularity containers on HPC to investigate multi-omics crosstalk (transcriptomics, genomics) between cancer cells and adipocytes, including data quality control (FastQC/MultiQC), alignment (STAR), quantification (Salmon), computational modelling of differential expression (DESeq2), and functional enrichment (KEGG/GO).
- Identified and validated candidate biomarker genes through gene enrichment analysis and gene ontology: in MM cells, SERF1A, CYP1A1, and GAREM1 were most downregulated while DHRSX, CFH, and IL10 were most upregulated; in BMAT cells, HYDIN, TRHDE, and JRK were most downregulated while CORO7-PAM16 and ITGAL were most upregulated.
- Characterized NF-κB and cytokine signalling dysregulation in multiple myeloma tumour-adipocyte interactions, revealing immune-mediated mechanisms with potential as drug discovery targets for therapeutic development.

Whole-Exome Sequencing: Variant Calling Pipeline (Human Genome hg38)

September 2024 – December 2024

- Executed a variant calling pipeline on whole-exome sequencing data aligned to the human reference genome (hg38), performing SNP and indel detection using GATK and bcftools with quality filtering and annotation.

Biostatistical Analysis of Multi-omics Data in Diabetes

September 2024 – December 2024

- Analyzed lipidomic and gene expression and proteomic-adjacent multi-omics data in R to investigate pancreatic beta-cell dysfunction during diabetes progression, designing data collection strategies for normalization, data transformation, and outlier removal to ensure statistical robustness.
- Built an elastic net regression model (glmnet) for feature selection and biomarker prioritization, and applied PCA and hierarchical clustering to identify lipidomic and transcriptomic patterns associated with beta-cell dysfunction.
- Performed correlation analyses with clinical traits (HbA1c, BMI) to prioritize candidate molecular features linked to disease progression.

EXPERIENCE

Pharmacy Assistant
Walmart Pharmacy

July 2026 – Present
Toronto, Canada

- Maintained data accuracy and SOP compliance processing 20+ prescriptions daily using Kroll and PharmaClik within a regulated Canadian dispensary environment.
- Applied systematic 7-step verification protocols across prescription records, patient data, and inventory audits, ensuring documentation integrity and quality standards through cross-functional collaboration with pharmacists, technicians, and healthcare stakeholders.

GMP Manufacturing & Process Technician
Rakesh Health Care India Limited

August 2023 – November 2023
Gandhinagar, India

- Led and trained a team of 15 associates within a regulated pharmaceutical manufacturing environment, streamlining workflows to improve productivity by 20% and reduce errors by 15%.
- Managed inventory and quality compliance in accordance with pharmaceutical regulatory standards, reducing material waste while improving overall production efficiency by 30%.
- Ensured adherence to GMP quality and production compliance standards, maintaining documentation integrity across packaging operations in a regulated pharma setting.

PROFESSIONAL CERTIFICATION

GLP (Good Laboratory Practices) Fundamentals, Biotalent Canada

January 2025

Python For Non-Programmer, LinkedIn Learning

July 2023

VOLUNTEER AND LEADERSHIP EXPERIENCE

Project Presenter, Northeastern University Presidential Visit

October 2025

Program Advisory Committee (PAC) Member, Northeastern University, Toronto

October 2024 – Present

Student Volunteer, MARS Discovery District, Toronto

May 2025